

Online Shoppers Purchasing Intention

Data Mining Project

Comprehensive Written Report

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Introduction

This is the complete data mining project based on a dataset of Online Shoppers Purchasing Intention. The project analyzes session level browsing behavior to see how it can be applied to real ecommerce decisions. The analysis covers: Data preparation, exploratory data analysis, feature engineering, regression modeling, classification modeling, K-Means clustering, association rule mining, recommendations and ethical considerations.

The project also has value as many e-commerce businesses want to recognize consumers who are likely to buy, consumers who require more assistance and sessions which are flagging abandonment. This project does not use just one specific data mining method to address these questions but instead employs multiple methods which address related questions in the realm of engagement, conversion, segmentation and behavioral patterns.

This dataset will be used to support students' learning in the following ways:

This dataset consists of 12,330 online shopping sessions and 18 attributes. Each row corresponds to a browsing session that includes data on pages visited, length of time on page, bounce rate, exit rate, page value, proximity to special day, month, operating system, browser, region, traffic type, visitor type, weekend status and whether the session ended in a purchase (Boolean Revenue).

The data set was selected as large enough to perform meaningful modelling and it has a mixture of numerical, categorical and Boolean variables. It also enables multiple data mining activities within a single consistent business context. Classification: The Revenue field is used to support classification. Regression: ProductRelated_Duration is used to support regression. The dataset also helps minimise the over-representation of a single user profile or campaign period as each session was targeted at a different user in a one-year period.

Dataset Attribute	Value
Original sessions	12,330
Cleaned sessions	12,205
Columns	18
Missing values	0
Duplicate rows removed	125
Purchase sessions	1,908
Non-purchase sessions	10,297
Cleaned purchase rate	15.6%

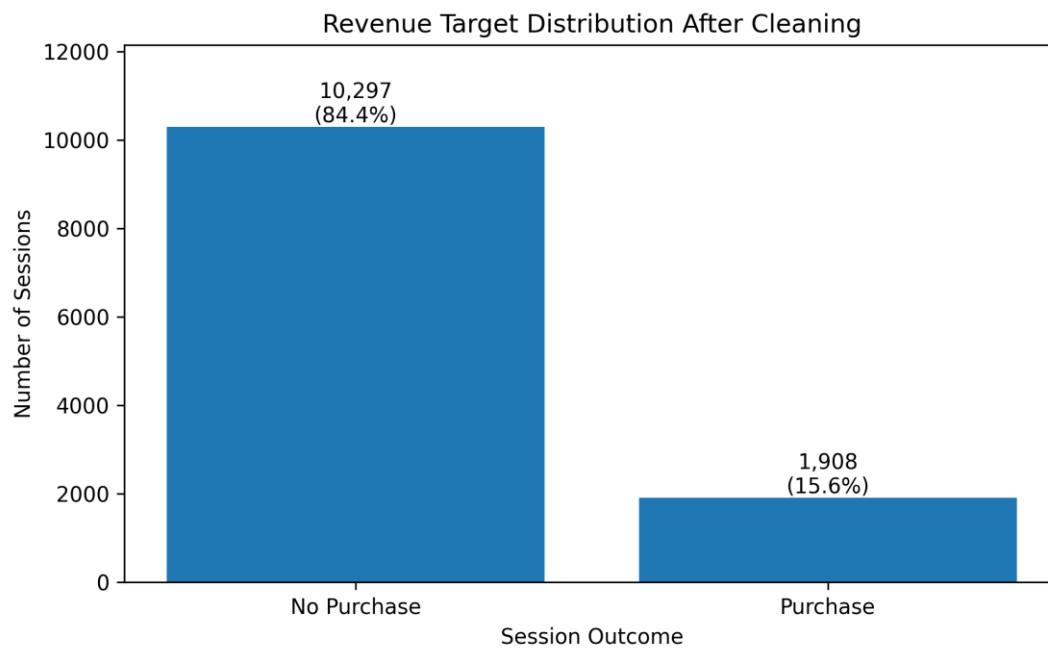


Figure 1. Purchase outcome distribution shows a strongly imbalanced target.

Data Preparation and Preprocessing

The data preparation started with missing value, duplicate row, data type, and class balance checks. No rows were deleted because they were exact duplicates in the original data set, and there were 0 rows that were removed due to missing values. After deduplication, there are 12,205 unique sessions. The target was imbalanced, as 15.6% of the cleaning sessions were of the Revenue = True class.

Common modelling mistakes were avoided by pre-processing. The median was used for imputing numeric features and they were standardized. The fields of operating system, browser, region, traffic type, visitor type, month, quarter and weekend were treated as categorical and one-hot encoded. It was significant that this was an identifier-like field because the operating system code 3 may not be in any meaningful behavioral sense higher than code 2.

Revenue and PageValues are omitted from the predictors for regression, as they can cause leakage of the outcome. Good for business interpretation but can be computed at the end of the session, so for the prediction goal, PageValues is a potential leakage risk.

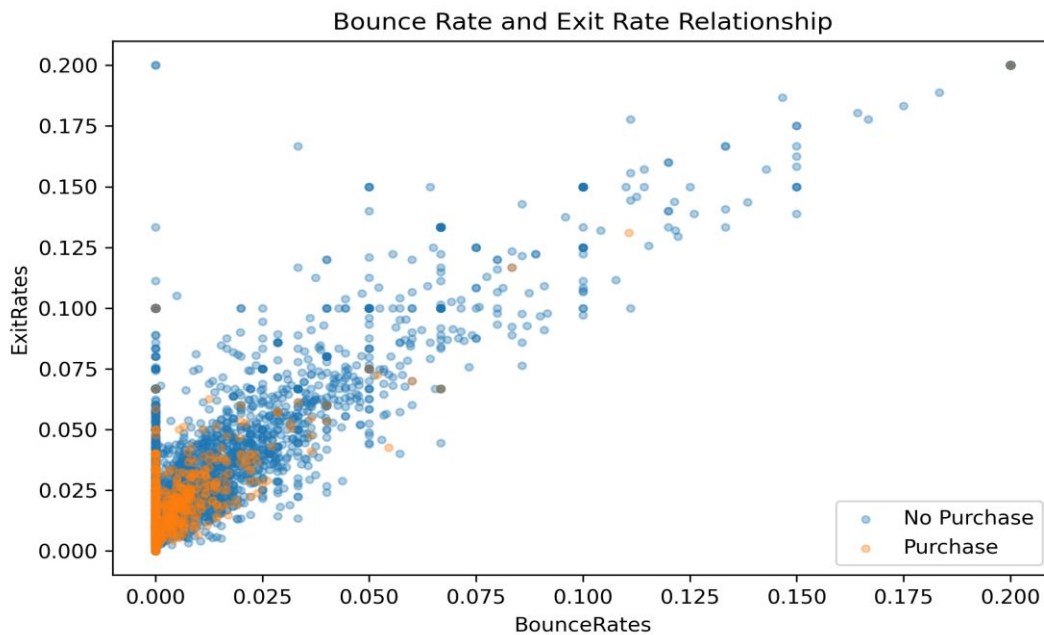


Figure 2. Bounce rate and exit rate show related but not identical abandonment behavior.

Exploratory Data Analysis

A study using EDA revealed that purchases do not occur often compared to non-purchasing sessions. Making the mistake of predicting the majority class would make accuracy an unsafe classification criterion, so this was a risky criterion to use. EDA also indicated that ProductRelated_Duration and PageValues are highly skewed – few sessions have very high engagement/payment values.

There were significant differences in conversion by visitor type and month. There were some differences in purchase behavior between new and returning visitors and November was a significant time of year. The results corroborate the recommendation to act with timing and context of the visitor rather than with a single action for all visits.

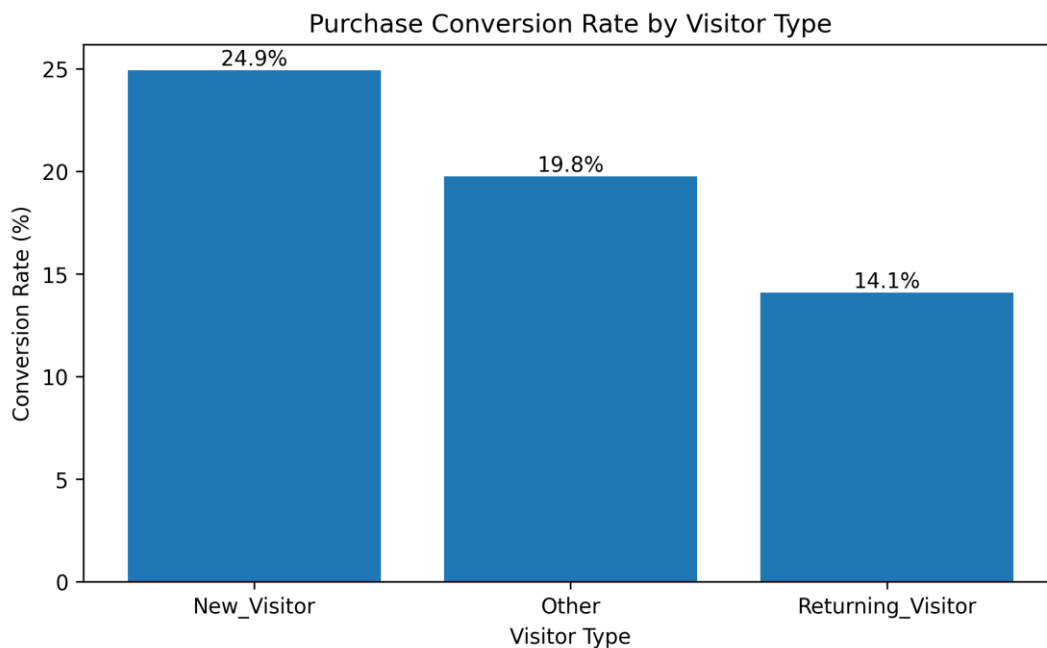


Figure 3. Conversion rate varies by visitor type.

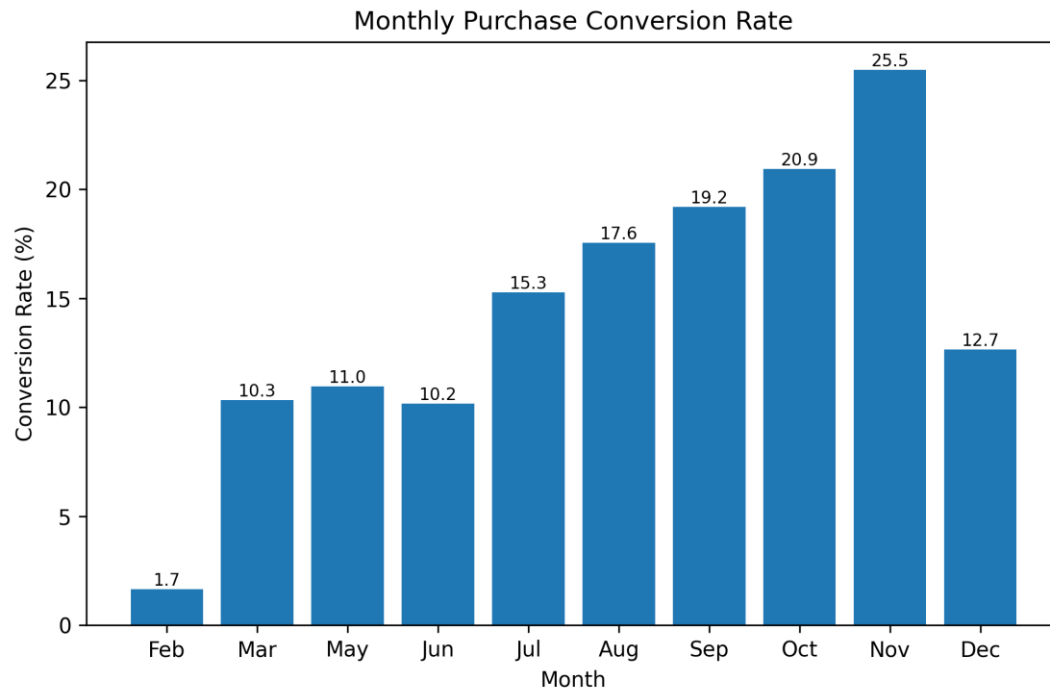


Figure 4. Monthly conversion patterns reveal seasonal behavior.

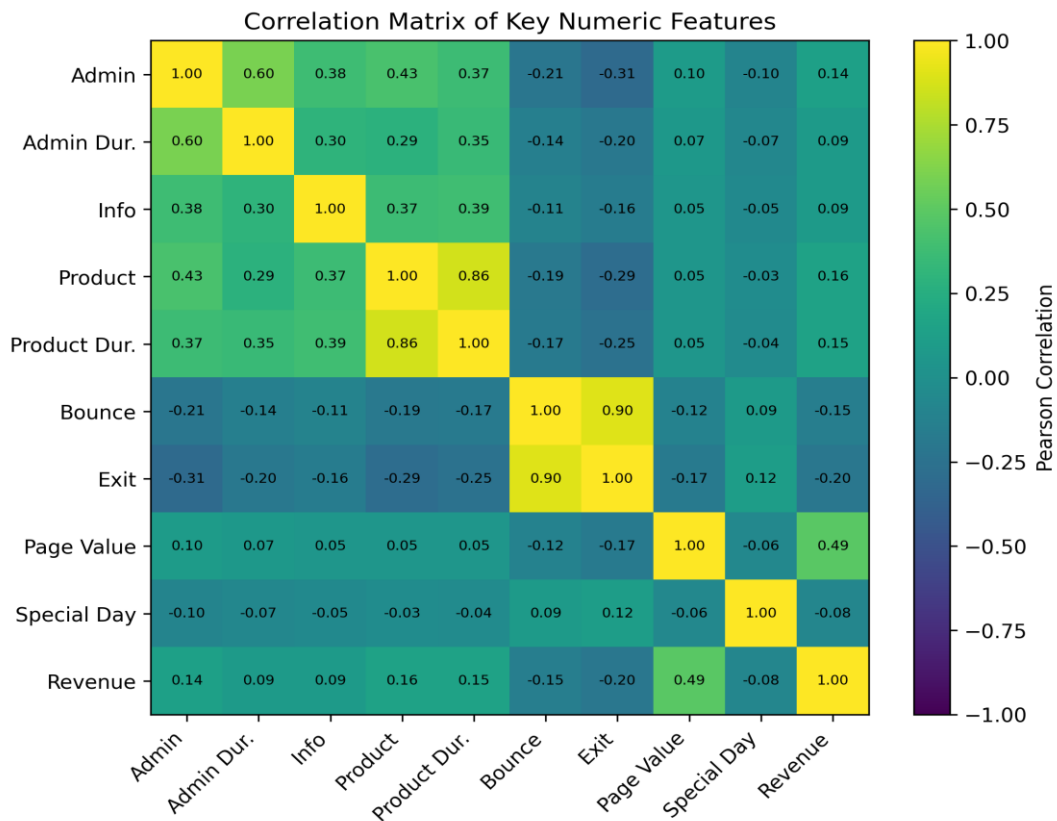


Figure 5. Correlations among selected behavioral variables and Revenue.

Feature Engineering

Feature engineering for raw variables representing user sessions to more meaningful variables representing user behavior. The session depth data was summarized in the TotalPageCount and TotalDuration metrics. NonProductPageCount and NonProductDuration isolated administrative and informational behavior from product browsing. Product engagement was assessed as intensity of product use (AvgProductTime). ProductPageShare was an indicator of the proportion of time spent on products during a session. EngagementScore integrated the duration with the decreased bounce and exit behavior. ExitBounceGap was a measure of the difference between two abandonment-related signals. SpecialDayWeekend was able to reveal the interaction between special day proximity and weekend activity. Time based features were offered by MonthNumber and Quarter, and the effect of extreme right skewness was mitigated by logTransform features.

These artificially created variables made it easier to interpret, as they represented more human behaviour than simply raw numbers of pages visited and how long they stayed on each page. They also facilitated clustering through increasing the sensitivity of behavioral differences.

Regression Modeling and Results

The regression task is an analysis of how long a visitor was on product-related pages, using a continuous measure of ProductRelated_Duration. This task is designed to address the question of how predictable browsing engagement is for a product, given the context of a session. Multiple Linear Regression, Ridge Regression and Random Forest Regression were evaluated by comparing the R-squared, MSE, RMSE and the cross validation.

Model	Test R-squared	Test MSE	Test RMSE	CV R-squared Mean	CV R-squared SD	CV MSE Mean	CV RMSE Mean	CV RMSE SD
Multiple Linear Regression	0.759	1,104,214.44	1,050.82	0.738	0.023	900,640.22	944.25	95.08
Ridge Regression	0.758	1,110,143.88	1,053.63	0.735	0.022	911,597.62	949.74	97.91
Random Forest Regression	0.705	1,353,660.22	1,163.47	0.773	0.033	787,568.08	878.80	123.59

Multiple Linear Regression had the lowest test RMSE (1,050.82) making it the best holdout regression model. This finding indicates that there is potential for modelling product page engagement with high accuracy, but extreme long-duration engagements are hard to predict. The residual plot shows that the errors increase when the browsing sessions become very long.

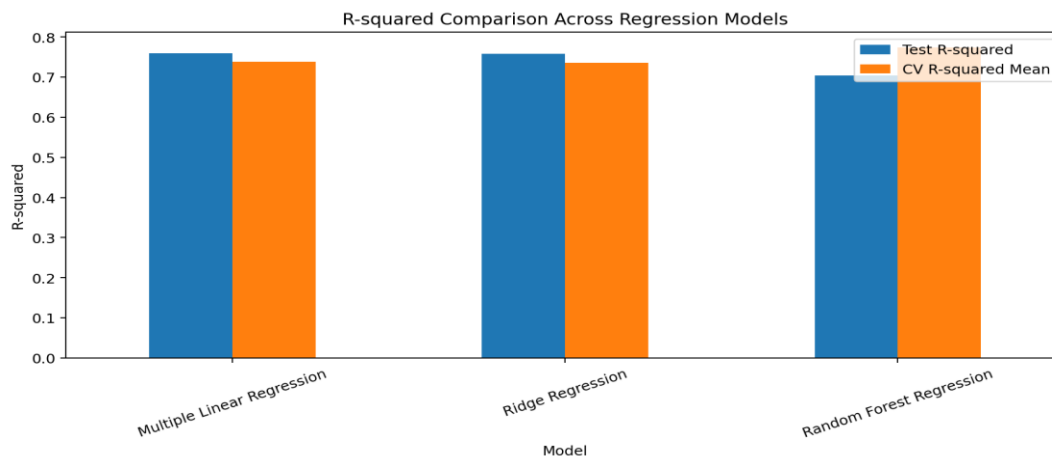


Figure 6. R-squared comparison for regression models.

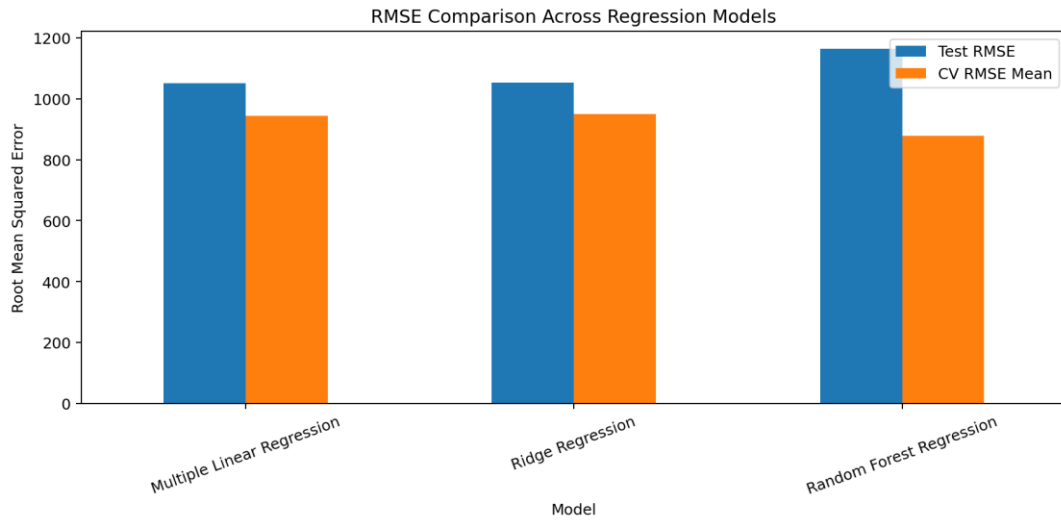


Figure 7. RMSE comparison for regression models.

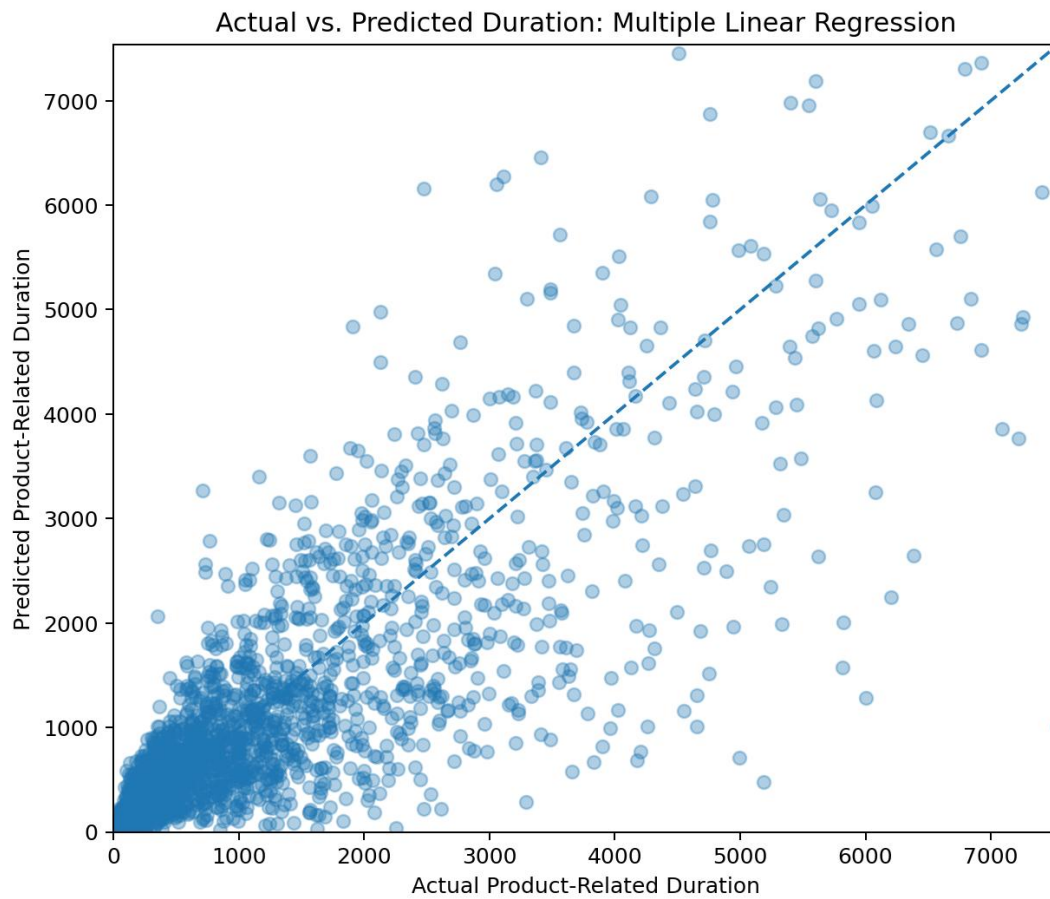


Figure 8. Actual versus predicted product-related duration.

Classification Modeling and Results

The classification task was to learn whether a session was followed by a purchase. Three models were tested: Decision Tree, Tuned Decision Tree and Random Forest. Target had class imbalance and the evaluation was based on F1 score and ROC-AUC as well as accuracy. To minimize the possibility of an incorrect majority class performance, the data was class-weighted and split into a training and testing set.

Model	Accuracy	F1 Score	ROC-AUC
Decision Tree	0.847	0.621	0.907
Tuned Decision Tree	0.828	0.615	0.927
Random Forest	0.895	0.597	0.924

The overall accuracy achieved by Random Forest is the highest, whereas the F1 score obtained by the Decision Tree is the best. The best ROC-AUC was obtained by the Tuned Decision Tree. The results demonstrate that the optimal model is dependent on the business goal. If the business appreciates that it needs to know more likely buyers, F1 or recall might be more important than accuracy. When the business is interested in ranking quality for interventions targeting, ROC-AUC is particularly useful.

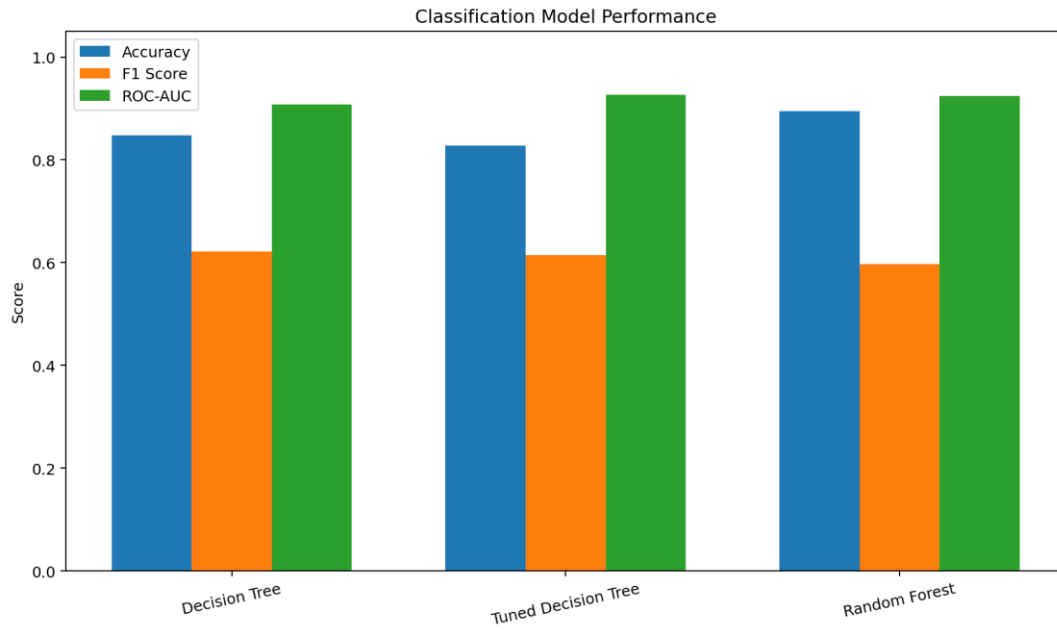


Figure 9. Classification model comparison.

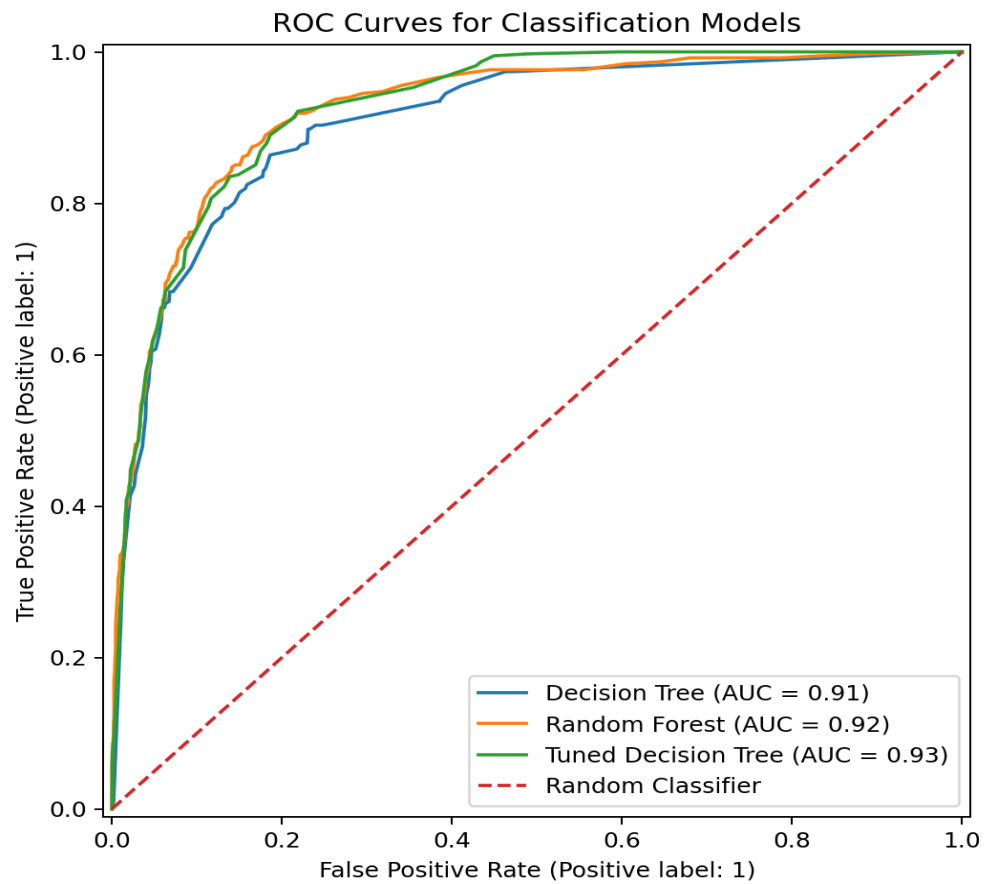


Figure 10. ROC curves compare ranking performance across classifiers.

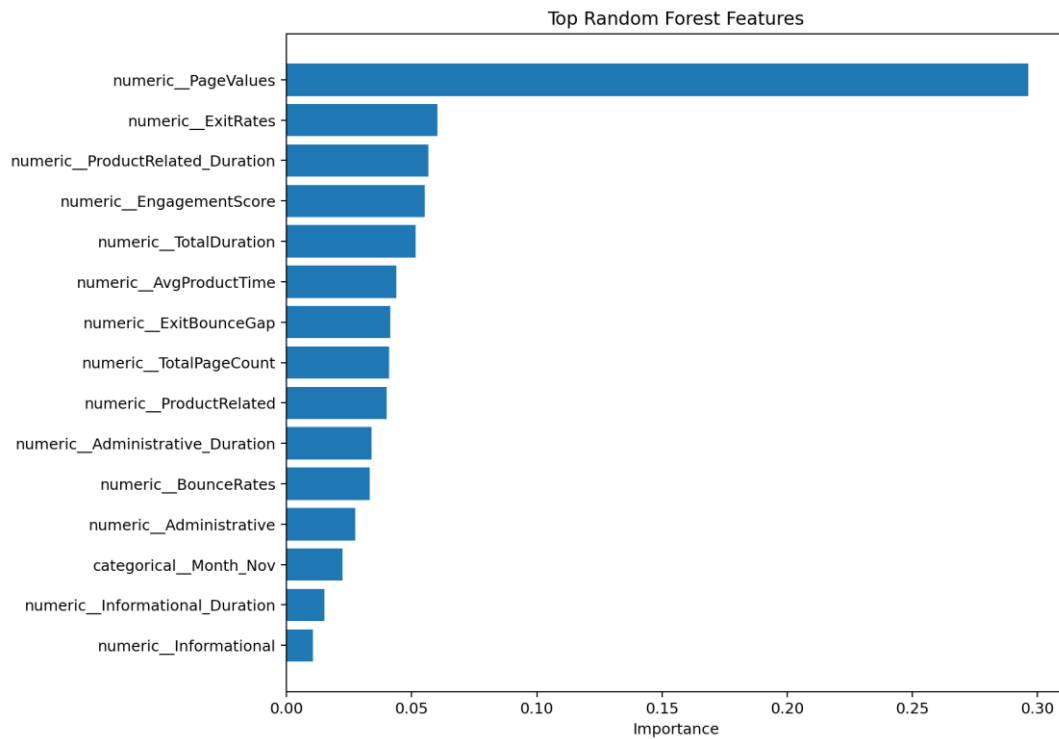


Figure 11. Random Forest feature importance highlights key purchase predictors.

Clustering Results

All behavior variables were log-transformed to reduce skewness and then standardized before being used in K-Means clustering. Rather than clustering the revenue, it was used after the clustering to interpret the outcome of the segments. The silhouette score of 3 clusters was 0.363, which was selected by silhouette analysis.

Cluster	Sessions	Purchase_Rate	Administrative_Pages	Informational_Pages	Product_Pages	Product_Duration	Bounce_Rate	Exit_Rate	Page_Value
Highly Engaged	2310	0.245	5.16	2.53	66.58	2,632.68	0.007	0.022	8.63

ed Resea rchers									
Typic al Produ ct Brows ers	902 1	0.148	1.84	0.039	26.07	955.18	0.010	0.033	5.84
Brief High- Bounc e Sessio ns	874	0.005	0.039	0.021	2.50	37.84	0.165	0.181	0.000

The three clusters were given the following names: Highly Engaged Researchers, Typical Product Browsers and Brief High-Bounce Sessions. The highest percentage of page visits and the greatest purchase rate was achieved by the Highly Engaged Researchers. Moderate behavior was represented by the Typical Product Browsers. Brief High-Bounce Sessions had a very low engagement rate and low rate of purchases, making them an important cohort to investigate their landing page, traffic source, and usability.

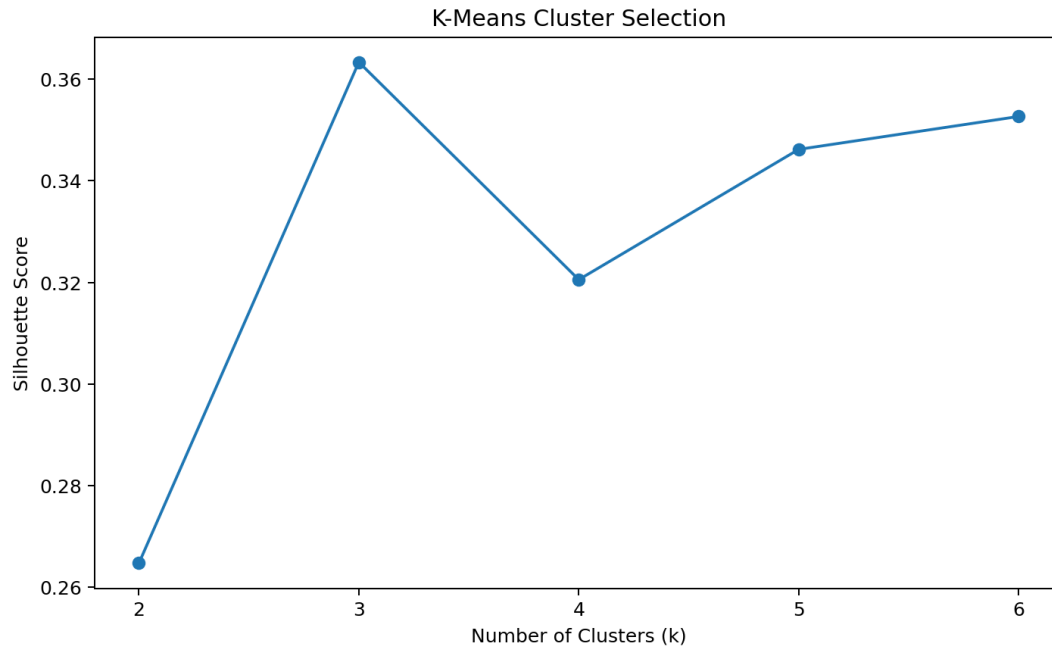


Figure 12. Silhouette scores used to select the number of clusters.

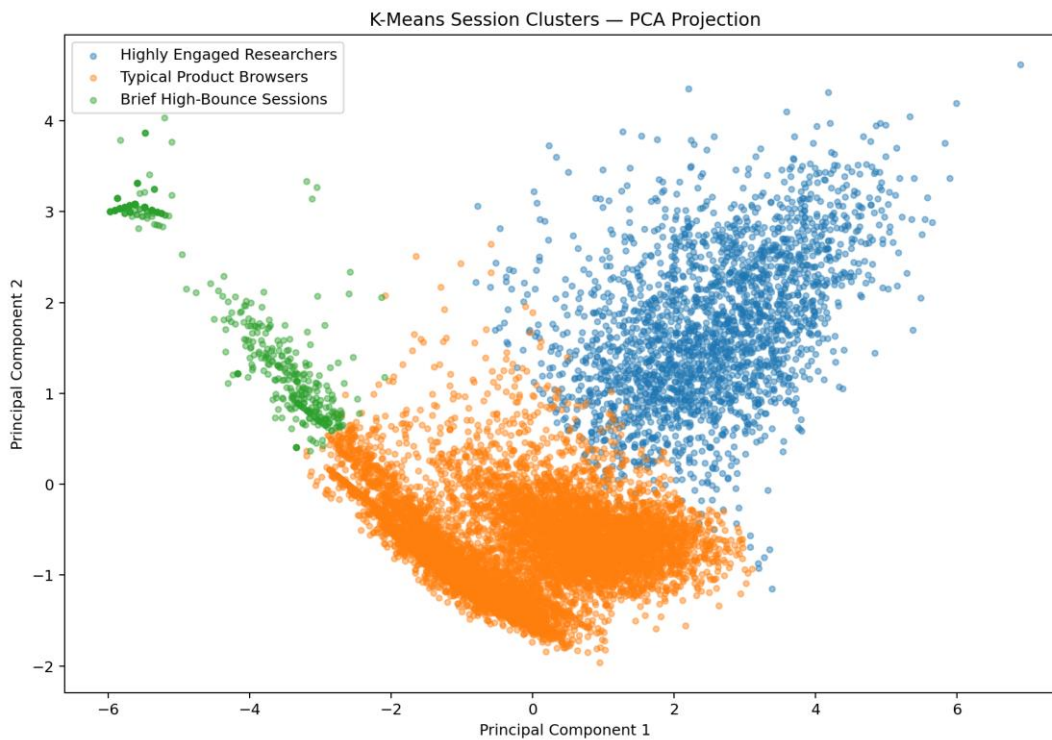


Figure 13. PCA projection of customer clusters.

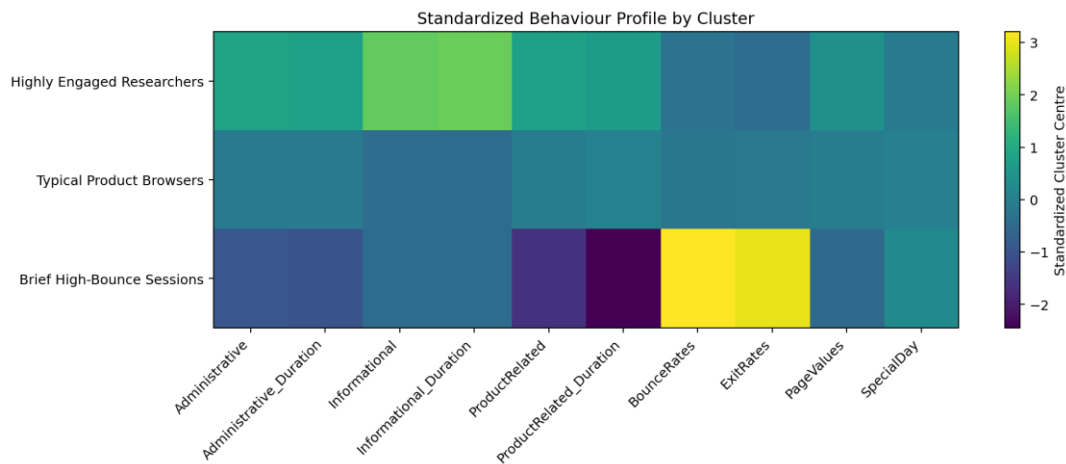


Figure 14. Cluster profile heatmap summarizes behavioral differences.

Association Rule Mining Results

Since this data set describes the session summary data instead of product baskets, association rule mining was adapted to session-level behavior items. The items included were positive page value, low bounce rate, low exit rate, high product engagement, month, weekend status, visitor type, cluster membership, and purchase outcome. Rules were understood as behavioral relations and not causal product-basket relations.

Antecedent	Consequent	Support	Confidence	Lift
Bounce_Low AND PageValue_High	Revenue_Yes	0.035	0.877	5.61
Cluster_Typical_Product_Browsers AND PageValue_High	Revenue_Yes	0.035	0.819	5.24
PageValue_High	Revenue_Yes	0.044	0.780	4.99
PageValue_High AND PageValue_Positive	Revenue_Yes	0.044	0.780	4.99
Bounce_Low AND	Revenue_Yes	0.071	0.732	4.68

PageValue_Positive				
Exit_Low AND PageValue_Positive	Revenue_Yes	0.060	0.672	4.30
Month_Nov AND PageValue_Positive	Revenue_Yes	0.042	0.659	4.22
Cluster_Typical_Product_Browsers AND PageValue_Positive	Revenue_Yes	0.087	0.631	4.04
PageValue_Positive AND Weekend	Revenue_Yes	0.033	0.583	3.73
PageValue_Positive	Revenue_Yes	0.126	0.563	3.60

Bounce_Low AND PageValue_High -> Purchase had the highest value of purchasing the rule with a confidence of 87.7% and a lift of 5.61. That is, sessions with that behavioral pattern were significantly more likely to make a purchase than the average session. These rules can be used to help inform trigger-based personalization or marketing prioritization but should be tested in controlled experiments before implementation in production.

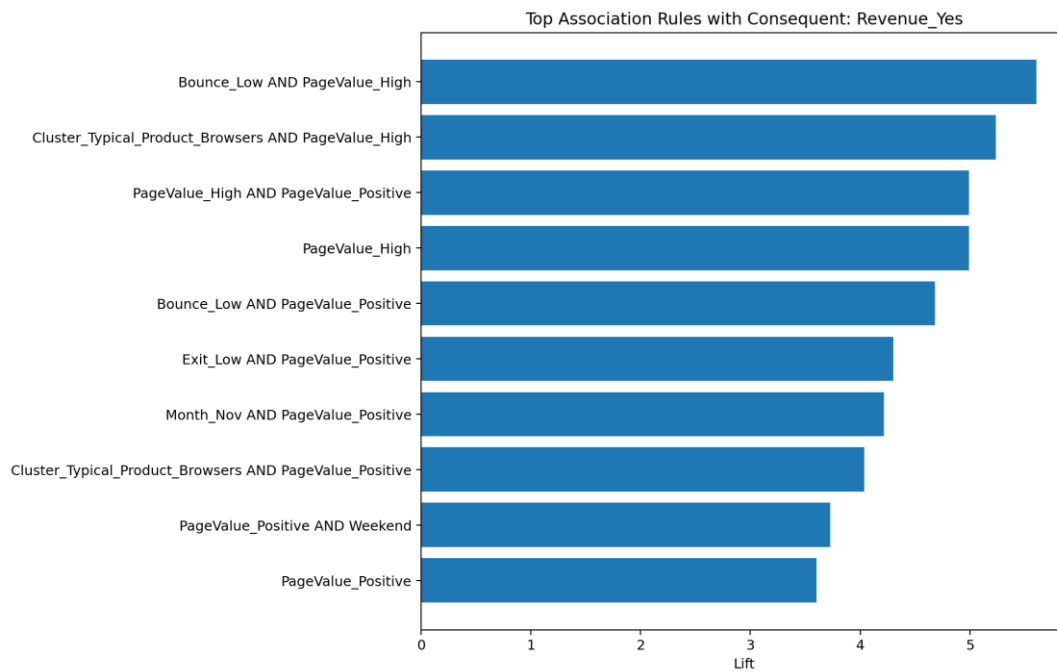


Figure 15. Top purchase-related association rules ranked by lift.

Practical Recommendations

- Optimize probability models to prioritize customer support, checkout reminders, or targeted offers, depending on the costs of false positives and false negatives.
- Enhance Highly Engaged Researchers' experience with comparison tools, in-depth product details, carts they save, reviews, and customized product recommendations.
- Enhance landing page relevance, page load speed, mobile usability and acquisition channel targeting for Brief High-Bounce Sessions.
- Focus on sessions with positive or high page value and low bounce behavior for timely interventions as these sessions are associated with good purchase.
- Take advantage of seasonal discoveries – particularly November activity – to schedule campaign resources, readiness, and checkout support.
- Take out the models after deployment and constantly watch for performance drift; adjust thresholds accordingly as customer usage patterns evolve.

Ethical Considerations

However, privacy, fairness, transparency, and the elimination of bias must be taken into consideration when using this project in an ethical way. While the data set provided does not contain explicit personal details like names or emails, it does contain information about how customers browse the Internet that could be sensitive. A real deployment is supposed to only capture what is needed, not store what isn't, provide secure access controls, and abide by relevant privacy policies and data retention policies.

Fairness is also a prerequisite. The information provided contains region, visitor type, traffic type, browser and operating system data which may be indicative of variations in customer access, device quality, geography and marketing exposure. A good model for the general population may not be good for smaller groups. The project addressed this concern by adopting strategies of stratified splitting, class weighting, multiple evaluation metrics, and discussion of subgroup monitoring. The production version should be able to report on false positive and false negative percentages by traffic source, region, device category, month, and visitor type.

Bias can also be historical business practices. The model can learn and reinforce patterns that are under-represented in historical data, if certain traffic channels, seasons, or user types are underrepresented. Therefore, the model should only be used for decision support and not as an automatic gatekeeper. Recommendations should be tested through A/B experiments, watched for unintended consequences, and documented, with clear boundaries of model limitations known by stakeholders.

What are some things that were hard and what can be improved in this?

The data set is at the session level, without page sequence, product identifiers, customer information, pricing, inventory and promotion information, or “real” product baskets. It restricts the ability to give reasons for buying or walking away. Association rules are not product co-purchase rules.

Additional future work involves supplementing with sequential clickstream data, progression through time on site, product categories, marketing campaign information, mobile performance metrics and real product transaction data.

In future models, gradient boosting, calibrated probability models, robust regression, threshold optimization, fairness audits, drift detection, and explainable AI approaches such as SHAP can be compared with each other. The most critical improvement might have been to determine the variables that could be used during the session and to create two models: one for the early session and one for the late session.

Conclusion

The project demonstrates meaningful signals in online shopping sessions, including predicting engagement, likelihood of purchase, customer segmentation, and identification of high-value patterns of behavior. Across analyses a number of page values, bounce behavior, exit behavior, product engagement, and engineered session quality measures were consistently important. The results justify the practical measures, including targeted support, enhancement of landing page quality, planning in advance for the season and responsible monitoring of the model. Privacy protection, fairness review, transparency, and bias and leakage considerations are elements of ethical deployment.

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